FireWatch_Indonesia_Report

September 9, 2026

1 FireWatch Indonesia

1.1 Analisis Spasial dan Temporal Titik Panas Kebakaran Hutan dan Lahan

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Author: Asrul
Periode Analisis: 1 Agustus - 7 September 2026 (37 hari)
Data Source: NASA FIRMS VIIRS 375m, NASA POWER API, GeoJSON Indonesia
Total Deteksi: 192,897 hotspot

1.2 Executive Summary

Proyek FireWatch Indonesia menganalisis pola persebaran titik panas kebakaran hutan dan lahan di seluruh Indonesia menggunakan data satelit NASA FIRMS, data cuaca NASA POWER, dan batas administrasi provinsi.

1.2.1 Key Findings:

1. **Kalimantan** merupakan wilayah paling parah (58.5% total hotspot)
2. **Kelembapan** adalah faktor cuaca terkuat ($r = -0.786$, $p < 0.001$)
3. **Puncak hotspot** terjadi pada 29 Agustus 2026 (11,100 deteksi)
4. **Hari panas** menghasilkan 84% lebih banyak hotspot dibanding hari normal

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2 1. Setup & Import Libraries

[90]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import Image, display
import warnings
warnings.filterwarnings('ignore')

plt.style.use('seaborn-v0_8-darkgrid')
sns.set_palette('husl')
```

```
print('Libraries loaded successfully!')
```

```
Libraries loaded successfully!
```

3 2. Dataset Overview

Analisis ini menggunakan tiga sumber data utama yang diintegrasikan untuk menghasilkan insights komprehensif.

3.1 2.1 NASA FIRMS - Hotspot Detection Data

Source: Fire Information for Resource Management System (FIRMS)
Sensor: VIIRS (Visible Infrared Imaging Radiometer Suite) 375m
Satellite: SNPP & NOAA-20
File: `data/raw/fire_nrt_SV-C2_800747.csv`

3.1.1 Dataset Characteristics

Attribute	Value
Total Records	192,897 hotspot detections
Temporal Coverage	1 Agustus - 7 September 2026 (37 hari)
Spatial Coverage	Seluruh wilayah Indonesia
Resolution	375 meter
Update Frequency	Near Real-Time (NRT)
Data Type	Point data (coordinates)

3.1.2 Key Variables

- **latitude, longitude**: Koordinat geografis deteksi (decimal degrees)
- **acq_date**: Tanggal akuisisi (YYYY-MM-DD)
- **acq_time**: Waktu akuisisi (HHMM UTC)
- **bright_ti4, bright_ti5**: Brightness temperature channel 4 & 5 (Kelvin)
- **frp**: Fire Radiative Power (Megawatts)
- **confidence**: Tingkat keyakinan deteksi (low/normal/high)
- **daynight**: Waktu deteksi (D=Day, N=Night)
- **satellite**: Satelit yang mendeteksi (N=SNPP, J1=NOAA-20)
- **instrument**: Instrumen sensor (VIIRS)
- **version**: Versi algoritma deteksi

3.1.3 Data Quality

- No missing values
- No duplicate records
- All coordinates within Indonesia boundaries
- 90.88% deteksi memiliki confidence level “normal”

3.2 2.2 NASA POWER - Weather Data

Source: Prediction Of Worldwide Energy Resources (POWER)
API: NASA POWER API v2.0
Files: - `data/raw/cuaca_indonesia_avg.csv` (daily national average) - `data/raw/cuaca_indonesia_detail.csv` (per-location detail)

3.2.1 Dataset Characteristics

Attribute	Value
Temporal Coverage	1 Agustus - 1 September 2026 (31 hari)
Spatial Coverage	32 lokasi sampling di Indonesia
Total Data Points	1,024 (32 locations × 32 days)
Data Type	Time series weather parameters
Aggregation	Daily average for national analysis

3.2.2 Sampling Strategy

Data cuaca dikumpulkan dari 32 titik koordinat yang tersebar di 8 region utama Indonesia: - **Sumatera**: 6 titik (Aceh, Riau, Lampung, Palembang, Jambi, Padang) - **Jawa**: 4 titik (Jakarta, Bandung, Semarang, Surabaya) - **Kalimantan**: 8 titik (Pontianak, Palangkaraya, Banjarmasin, Samarinda, dll) - **Sulawesi**: 4 titik (Makassar, Manado, Palu, Kendari) - **Papua**: 4 titik (Jayapura, Sorong, Timika, Merauke) - **Maluku**: 2 titik (Ambon, Ternate) - **Nusa Tenggara**: 3 titik (Denpasar, Mataram, Kupang) - **Bali**: 1 titik (Denpasar)

3.2.3 Weather Parameters

- **T2M_MAX**: Suhu maksimum harian at 2 meters (°C)
- **PRECTOTCORR**: Curah hujan terkoreksi (mm/hari)
- **RH2M**: Kelembapan relatif rata-rata at 2 meters (%)

3.2.4 Data Processing

1. Download data per lokasi via NASA POWER API
2. Calculate daily national average dari 32 lokasi
3. Merge dengan daily hotspot counts berdasarkan tanggal
4. Handle missing values dengan interpolation jika diperlukan

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3.3 2.3 Indonesia Administrative Boundaries

Source: GeoJSON Indonesia Province Boundaries
Format: GeoJSON (Geographic JSON)
Files: - `data/raw/indonesia-province-simple.json` (simplified geometry) - `data/raw/id-all.geo.json` (detailed geometry)

3.3.1 Dataset Characteristics

Attribute	Value
Level	Province (Provinsi)
Total Polygons	34 provinsi Indonesia
Coordinate System	WGS84 (EPSG:4326)
Format	GeoJSON with properties
Geometry Type	MultiPolygon

3.3.2 Usage in Analysis

- **Spatial Join**: Mapping hotspot points ke provinsi boundaries
- **Choropleth Map**: Visualisasi jumlah hotspot per provinsi
- **Overlay Analysis**: Kombinasi point dan polygon layers
- **Aggregation**: Summary statistics per provinsi

3.3.3 Provinces Covered

Seluruh 34 provinsi Indonesia termasuk: - 10 provinsi di Sumatera - 6 provinsi di Jawa - 5 provinsi di Kalimantan - 6 provinsi di Sulawesi - 5 provinsi di Papua - 2 provinsi di Bali & Nusa Tenggara

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3.4 2.4 Data Integration Workflow

3.4.1 Step 1: Spatial Integration

- Load hotspot points (192,897 records)
- Load province boundaries (34 provinces)
- Perform spatial join: Point-in-Polygon
- Result: `data/processed/hotspot_per_provinsi.csv`
- Mapped: 191,697 hotspots (99.4% success rate)

3.4.2 Step 2: Temporal Integration

- Aggregate hotspot counts per day (37 days)
- Calculate daily weather averages (31 days)
- Merge by date
- Result: `data/processed/weather_hotspot_merged.csv`
- Valid days: 29 (after removing missing weather data)

3.4.3 Step 3: Analysis Outputs

Generated processed files: - `data_inspection_summary.txt`: Quality check results - `eda_summary.txt`: Exploratory analysis findings - `spatial_analysis_summary.txt`: Province-level statistics - `weather_hotspot_summary.txt`: Correlation analysis results

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3.5 2.5 Data Limitations

3.5.1 Hotspot Data

- Satellite detection limited by cloud cover
- 375m resolution may miss small fires
- Hotspot fire area (just detection point)
- Industrial heat sources may cause false positives
- Night detection capability lower than daytime

3.5.2 Weather Data

- Point measurements may not represent all conditions
- National average smooths local variations
- Weather-fire relationship has time lag
- Limited to 31 days (shorter than hotspot data)

3.5.3 Boundary Data

- Province-level may hide district variations
- Political boundaries may not match ecological zones
- Coastal areas have complex geometries

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[120]:
```
# Load hotspot dataset
df = pd.read_csv('../data/raw/fire_nrt_SV-C2_800747.csv')
df['acq_date'] = pd.to_datetime(df['acq_date'])  # Convert to datetime

print(f'Dataset Shape: {df.shape}')
print(f'Total Hotspot: {len(df):,}')
print(f'\nColumns: {list(df.columns)}')
print(f'\nData Types:')
print(df.dtypes)
```

```
Dataset Shape: (192897, 14)
Total Hotspot: 192,897

Columns: ['latitude', 'longitude', 'brightness', 'scan', 'track', 'acq_date',
'acq_time', 'satellite', 'instrument', 'confidence', 'version', 'bright_t31',
'frp', 'daynight']

Data Types:
latitude             float64
longitude            float64
brightness           float64
scan                 float64
track                float64
acq_date      datetime64[ns]
acq_time               int64
satellite             object
instrument            object
confidence            object
version               object
```

```
bright_t31           float64
frp                  float64
daynight              object
dtype: object
```

[92]:
```
# First 5 rows
df.head()
```

[92]:

	latitude	longitude	brightness	scan	track	acq_date	acq_time	\
0	-7.82814	139.04887	338.59	0.43	0.46	2026-08-01	402	
1	-8.06223	140.56680	348.05	0.52	0.42	2026-08-01	402	
2	-8.06363	140.55728	332.29	0.52	0.42	2026-08-01	402	
3	-8.07347	140.49045	328.67	0.53	0.42	2026-08-01	402	
4	-8.05176	140.58467	329.44	0.52	0.41	2026-08-01	402	

	satellite	instrument	confidence	version	bright_t31	frp	daynight
0	SNPP	SNPP	n	2.0NRT	296.55	9.25	D
1	SNPP	SNPP	n	2.0NRT	298.69	18.24	D
2	SNPP	SNPP	n	2.0NRT	296.66	4.26	D
3	SNPP	SNPP	l	2.0NRT	299.51	6.33	D
4	SNPP	SNPP	n	2.0NRT	296.81	5.05	D

[93]:
```
# Data quality summary
print('='*70)
print('DATA QUALITY CHECK')
print('='*70)
print(f'Total Records      : {len(df):,}')
print(f'Total Columns      : {df.shape[1]}')
print(f'Missing Values     : {df.isnull().sum().sum()}')
print(f'Duplicate Records  : {df.duplicated().sum()}')
print(f'Date Range         : {df["acq_date"].min()} to {df["acq_date"].max()}')
print(f'Latitude Range     : {df["latitude"].min():.2f}° to {df["latitude"].max():.
 ↪2f}°')
print(f'Longitude Range    : {df["longitude"].min():.2f}° to {df["longitude"].max():.
 ↪2f}°')
print('='*70)
print('\nData quality: PASSED - No missing values, no duplicates!')
```

```
======================================================================
DATA QUALITY CHECK
======================================================================
Total Records      : 192,897
Total Columns      : 14
Missing Values     : 0
Duplicate Records  : 0
Date Range         : 2026-08-01 00:00:00 to 2026-09-07 00:00:00
Latitude Range     : -10.85° to 5.60°
Longitude Range    : 95.34° to 140.98°
======================================================================

Data quality: PASSED - No missing values, no duplicates!
```

4 3. Exploratory Data Analysis

4.1 3.1 Analisis Temporal

4.1.1 Tren Harian Hotspot

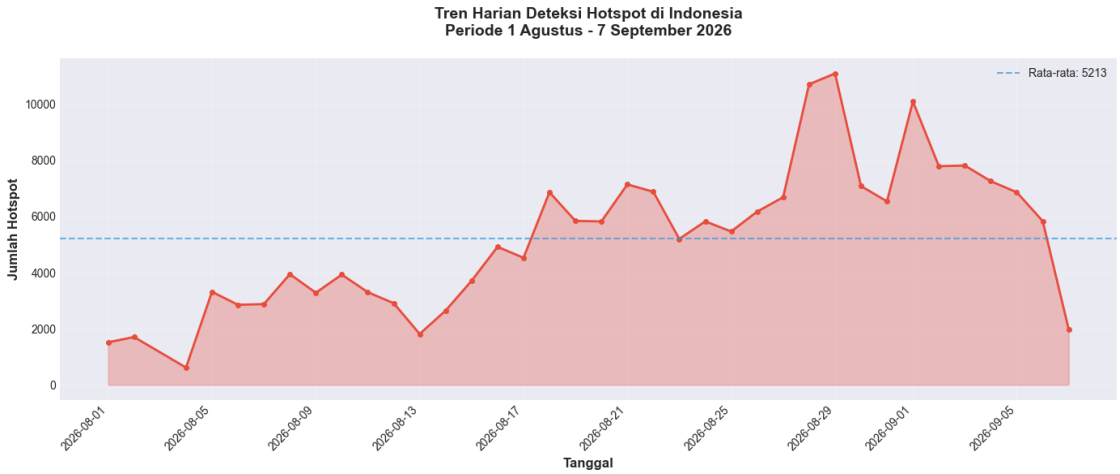
[94]:
```
# Tren Harian
daily_hotspot = df.groupby('acq_date').size().reset_index(name='jumlah')

mean_val = daily_hotspot['jumlah'].mean()
max_val = daily_hotspot['jumlah'].max()
min_val = daily_hotspot['jumlah'].min()
max_date = daily_hotspot.loc[daily_hotspot['jumlah'].idxmax(), 'acq_date']
min_date = daily_hotspot.loc[daily_hotspot['jumlah'].idxmin(), 'acq_date']

print(f"Rata-rata hotspot/hari: {mean_val:.0f}")
print(f"Hotspot tertinggi: {max_val:,} pada {max_date}")
print(f"Hotspot terendah: {min_val:,} pada {min_date}")

plt.figure(figsize=(14, 6))
plt.plot(daily_hotspot['acq_date'], daily_hotspot['jumlah'],
         marker='o', linewidth=2, markersize=4, color='#e74c3c')
plt.axhline(y=mean_val, color='#3498db', linestyle='--',
            linewidth=1.5, label=f'Rata-rata: {mean_val:.0f}', alpha=0.7)
plt.fill_between(daily_hotspot['acq_date'], daily_hotspot['jumlah'], alpha=0.3,␣
 ↪color='#e74c3c')
plt.xlabel('Tanggal', fontsize=12, fontweight='bold')
plt.ylabel('Jumlah Hotspot', fontsize=12, fontweight='bold')
plt.title('Tren Harian Deteksi Hotspot di Indonesia\nPeriode 1 Agustus - 7 September␣
 ↪2026',
          fontsize=14, fontweight='bold', pad=20)
plt.xticks(rotation=45, ha='right')
plt.legend(loc='upper right')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

```
Rata-rata hotspot/hari: 5213
Hotspot tertinggi: 11,100 pada 2026-08-29 00:00:00
Hotspot terendah: 615 pada 2026-08-04 00:00:00
```



Key Findings:

- **Rata-rata:** 5,213 hotspot per hari
- **Puncak:** 11,100 hotspot pada 29 Agustus 2026
- **Terendah:** 615 hotspot pada 4 Agustus 2026
- **Tren:** Peningkatan signifikan pada akhir Agustus

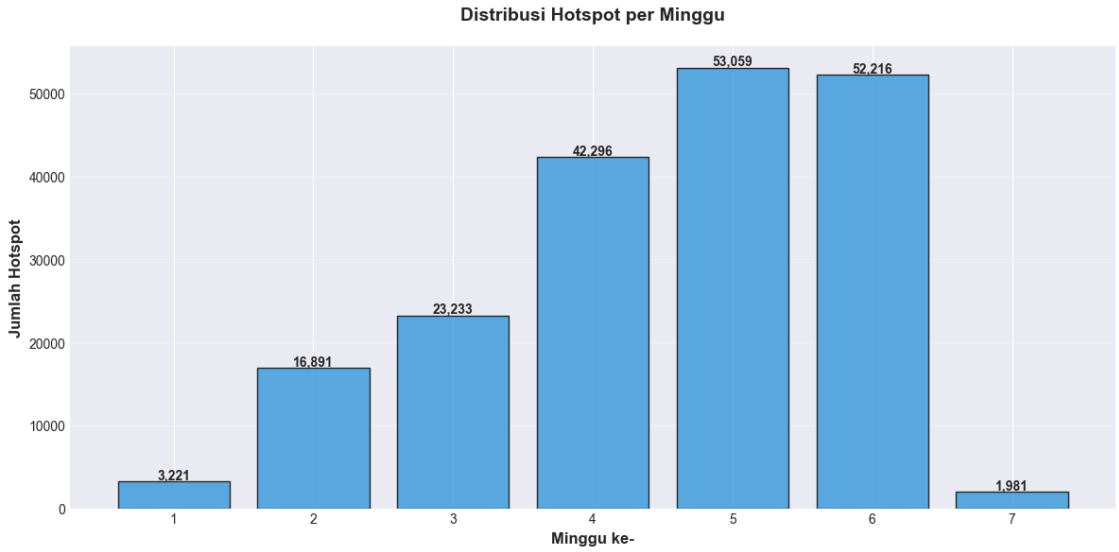
4.1.2 Tren Mingguan

[95]:
```
# Tren Mingguan
daily_hotspot['week'] = daily_hotspot['acq_date'].dt.isocalendar().week
weekly_hotspot = daily_hotspot.groupby('week')['jumlah'].sum().reset_index()

plt.figure(figsize=(12, 6))
bars = plt.bar(range(1, len(weekly_hotspot)+1), weekly_hotspot['jumlah'],
               color='#3498db', alpha=0.8, edgecolor='black')

for i, bar in enumerate(bars):
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2., height,
             f'{int(height):,}', ha='center', va='bottom', fontsize=10,␣
 ↪fontweight='bold')

plt.xlabel('Minggu ke-', fontsize=12, fontweight='bold')
plt.ylabel('Jumlah Hotspot', fontsize=12, fontweight='bold')
plt.title('Distribusi Hotspot per Minggu', fontsize=14, fontweight='bold', pad=20)
plt.xticks(range(1, len(weekly_hotspot)+1))
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()
```



4.2 3.2 Karakteristik Hotspot

4.2.1 Confidence Level Distribution

[96]:
```
# Confidence Level Distribution
confidence_dist = df['confidence'].value_counts().sort_index()
confidence_pct = (confidence_dist / len(df) * 100).round(2)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

# Bar chart
colors_conf = ['#f39c12', '#3498db', '#e74c3c']
bars = ax1.bar(confidence_dist.index, confidence_dist.values, color=colors_conf,
               edgecolor='black', linewidth=1.5)
for bar in bars:
    height = bar.get_height()
    ax1.text(bar.get_x() + bar.get_width()/2., height,
            f'{int(height):,}\n({height/len(df)*100:.1f}%)',
            ha='center', va='bottom', fontsize=10, fontweight='bold')

ax1.set_xlabel('Confidence Level', fontsize=12, fontweight='bold')
ax1.set_ylabel('Jumlah Deteksi', fontsize=12, fontweight='bold')
ax1.set_title('Distribusi Confidence Level', fontsize=13, fontweight='bold')
ax1.grid(True, alpha=0.3, axis='y')

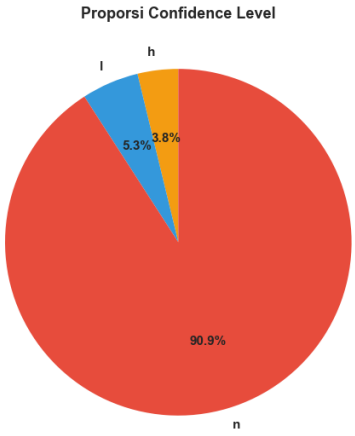
# Pie chart
ax2.pie(confidence_dist.values, labels=confidence_dist.index, autopct='%1.1f%%',
        colors=colors_conf, startangle=90,
        textprops={'fontsize': 11, 'fontweight': 'bold'})
ax2.set_title('Proporsi Confidence Level', fontsize=13, fontweight='bold')

plt.tight_layout()
```

```
plt.show()

print('Distribusi Confidence Level:')
for level in confidence_dist.index:
    print(f'  {level}: {confidence_dist[level]:,} ({confidence_pct[level]:.2f}%)')
```






```
Distribusi Confidence Level:
  h: 7,319 (3.79%)
  l: 10,279 (5.33%)
  n: 175,299 (90.88%)
```

- **Normal:** 90.88% (175,299 deteksi)
- **Low:** 5.33% (10,279 deteksi)
- **High:** 3.79% (7,319 deteksi)

Mayoritas deteksi memiliki confidence level normal, menunjukkan kualitas data yang reliable.

4.2.2 Deteksi Siang vs Malam

[97]:
```
# Day vs Night
dn_dist = df['daynight'].value_counts()
dn_labels = {'D': 'Day (Siang)', 'N': 'Night (Malam)'}

print('Distribusi Siang vs Malam:')
for dn in ['D', 'N']:
    if dn in dn_dist.index:
        count = dn_dist[dn]
        pct = (count/len(df))*100
        print(f"  {dn_labels[dn]:15s}: {count:7,} ({pct:5.2f}%)")

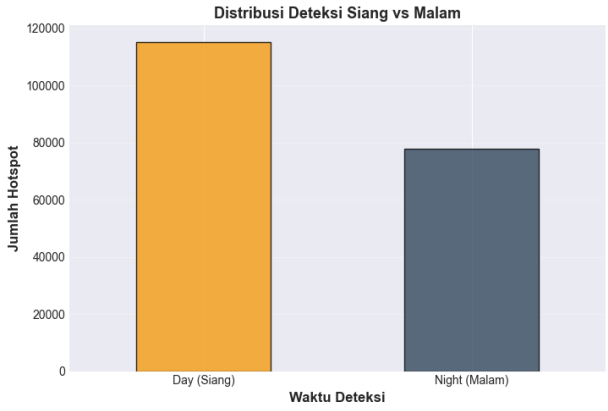
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
```

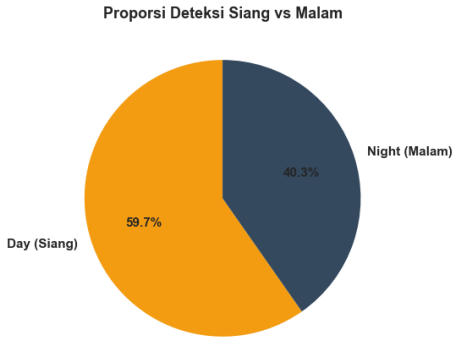
```
# Bar chart
dn_plot = dn_dist.copy()
dn_plot.index = dn_plot.index.map(dn_labels)
dn_plot.plot(kind='bar', ax=ax1, color=['#f39c12', '#34495e'], alpha=0.8,␣
 ↪edgecolor='black')
ax1.set_xlabel('Waktu Deteksi', fontsize=12, fontweight='bold')
ax1.set_ylabel('Jumlah Hotspot', fontsize=12, fontweight='bold')
ax1.set_title('Distribusi Deteksi Siang vs Malam', fontsize=13, fontweight='bold')
ax1.tick_params(axis='x', rotation=0)
ax1.grid(axis='y', alpha=0.3)

# Pie chart
ax2.pie(dn_plot.values, labels=dn_plot.index, autopct='%1.1f%%',
        startangle=90, colors=['#f39c12', '#34495e'], textprops={'fontsize': 11,␣
 ↪'fontweight': 'bold'})
ax2.set_title('Proporsi Deteksi Siang vs Malam', fontsize=13, fontweight='bold')

plt.tight_layout()
plt.show()
```

```
Distribusi Siang vs Malam:
  Day (Siang)    : 115,217 (59.73%)
  Night (Malam)  :  77,680 (40.27%)
```





- **Siang (Day):** 59.73% (115,217 deteksi)
- **Malam (Night):** 40.27% (77,680 deteksi)

Deteksi lebih banyak terjadi di siang hari, kemungkinan karena aktivitas pembakaran lahan dan kondisi cuaca yang lebih panas.

4.2.3 Brightness Temperature & Fire Radiative Power

[98]:
```
# Brightness & FRP
print(f"Brightness - Mean: {df['brightness'].mean():.2f}, Median: {df['brightness'].
 ↪median():.2f}")
print(f"FRP - Mean: {df['frp'].mean():.2f}, Median: {df['frp'].median():.2f}")

fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# Brightness distribution
axes[0,0].hist(df['brightness'], bins=50, color='#e74c3c', alpha=0.7,␣
 ↪edgecolor='black')
axes[0,0].axvline(df['brightness'].mean(), color='blue', linestyle='--', linewidth=2,
                  label=f'Mean: {df["brightness"].mean():.1f}K')
axes[0,0].axvline(df['brightness'].median(), color='green', linestyle='--',␣
 ↪linewidth=2,
                  label=f'Median: {df["brightness"].median():.1f}K')
axes[0,0].set_xlabel('Brightness (K)', fontweight='bold')
axes[0,0].set_ylabel('Frequency', fontweight='bold')
axes[0,0].set_title('Distribusi Brightness', fontweight='bold')
axes[0,0].legend()
axes[0,0].grid(alpha=0.3)

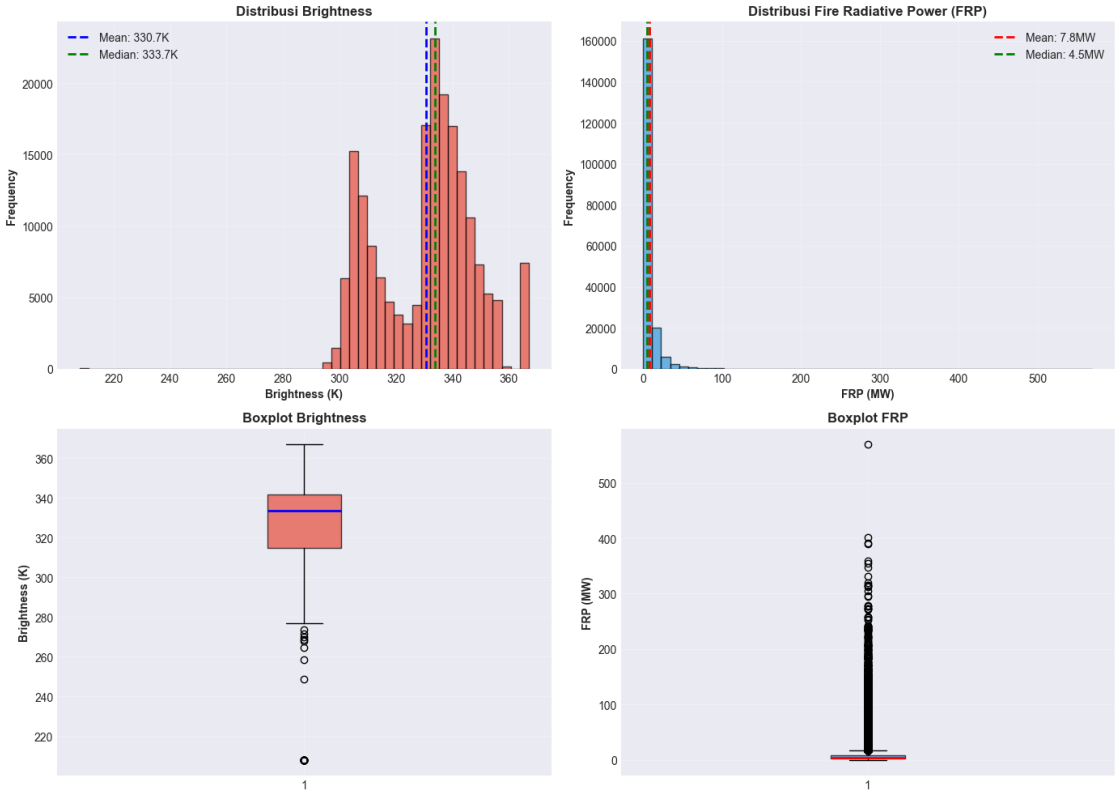
# FRP distribution
axes[0,1].hist(df['frp'], bins=50, color='#3498db', alpha=0.7, edgecolor='black')
axes[0,1].axvline(df['frp'].mean(), color='red', linestyle='--', linewidth=2,
                  label=f'Mean: {df["frp"].mean():.1f}MW')
axes[0,1].axvline(df['frp'].median(), color='green', linestyle='--', linewidth=2,
                  label=f'Median: {df["frp"].median():.1f}MW')
axes[0,1].set_xlabel('FRP (MW)', fontweight='bold')
axes[0,1].set_ylabel('Frequency', fontweight='bold')
axes[0,1].set_title('Distribusi Fire Radiative Power (FRP)', fontweight='bold')
axes[0,1].legend()
axes[0,1].grid(alpha=0.3)

# Brightness boxplot
axes[1,0].boxplot(df['brightness'], vert=True, patch_artist=True,
                  boxprops=dict(facecolor='#e74c3c', alpha=0.7),
                  medianprops=dict(color='blue', linewidth=2))
axes[1,0].set_ylabel('Brightness (K)', fontweight='bold')
axes[1,0].set_title('Boxplot Brightness', fontweight='bold')
axes[1,0].grid(alpha=0.3)

# FRP boxplot
axes[1,1].boxplot(df['frp'], vert=True, patch_artist=True,
                  boxprops=dict(facecolor='#3498db', alpha=0.7),
                  medianprops=dict(color='red', linewidth=2))
axes[1,1].set_ylabel('FRP (MW)', fontweight='bold')
axes[1,1].set_title('Boxplot FRP', fontweight='bold')
axes[1,1].grid(alpha=0.3)

plt.tight_layout()
plt.show()
```

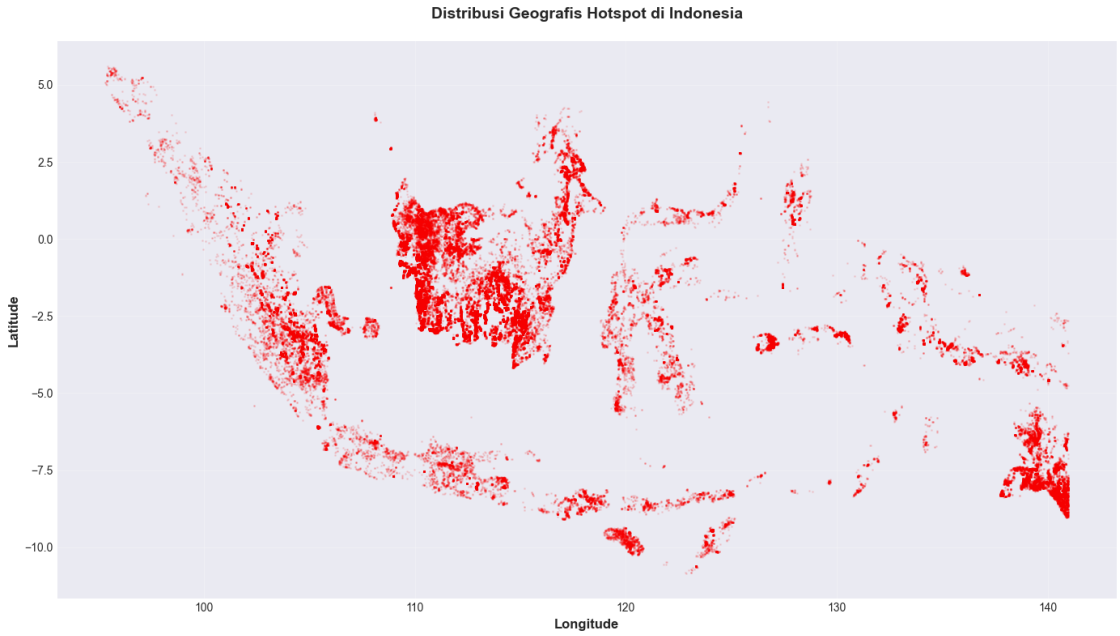
```
Brightness - Mean: 330.70, Median: 333.70
FRP - Mean: 7.83, Median: 4.48
```



4.3 3.3 Distribusi Geografis

4.3.1 Peta Persebaran Hotspot

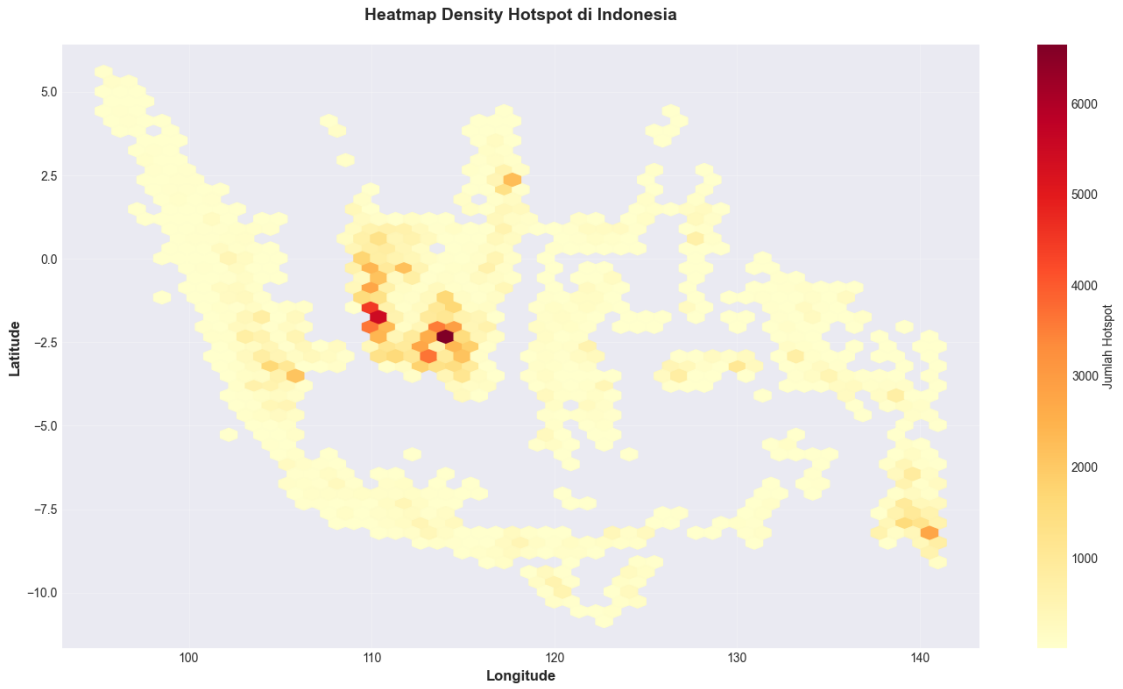
[99]:
```
# Distribusi Geografis
plt.figure(figsize=(14, 8))
plt.scatter(df['longitude'], df['latitude'], alpha=0.1, s=1, c='red')
plt.xlabel('Longitude', fontsize=12, fontweight='bold')
plt.ylabel('Latitude', fontsize=12, fontweight='bold')
plt.title('Distribusi Geografis Hotspot di Indonesia', fontsize=14, fontweight='bold',␣
↪pad=20)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Hotspot tersebar di seluruh Indonesia dengan konsentrasi tertinggi di Kalimantan Barat & Tengah dan Papua.

4.3.2 Heatmap Density

[104]:
```
# Heatmap Density
plt.figure(figsize=(14, 8))
plt.hexbin(df['longitude'], df['latitude'], gridsize=50, cmap='YlOrRd', mincnt=1)
plt.colorbar(label='Jumlah Hotspot')
plt.xlabel('Longitude', fontsize=12, fontweight='bold')
plt.ylabel('Latitude', fontsize=12, fontweight='bold')
plt.title('Heatmap Density Hotspot di Indonesia', fontsize=14, fontweight='bold',␣
↪pad=20)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



5 4. Spatial Analysis per Provinsi

Analisis spatial menggunakan GeoPandas untuk melakukan spatial join antara hotspot points dengan province boundaries.

5.0.1 Top 10 Provinsi

[105]:
```
# Spatial Analysis - Load and Process Province Data
import geopandas as gpd
from shapely.geometry import Point

# Load province boundaries
provinces = gpd.read_file('../data/raw/indonesia-province-simple.json')

# Convert hotspot to GeoDataFrame
geometry = [Point(xy) for xy in zip(df['longitude'], df['latitude'])]
hotspot_gdf = gpd.GeoDataFrame(df, geometry=geometry, crs='EPSG:4326')

# Spatial join
hotspot_provinces = gpd.sjoin(hotspot_gdf, provinces, how='left', predicate='within')

# Count per province
province_counts = hotspot_provinces.groupby('Propinsi').size().
↪reset_index(name='hotspot_count')
province_counts = province_counts.sort_values('hotspot_count', ascending=False)
```

```
province_counts['percentage'] = (province_counts['hotspot_count'] / len(df) * 100).
 ↪round(2)

# Merge with geometries for mapping
provinces_merged = provinces.merge(province_counts, on='Propinsi', how='left')
provinces_merged['hotspot_count'] = provinces_merged['hotspot_count'].fillna(0)

print(f'Total provinces: {len(provinces)}')
print(f'Provinces with hotspot: {len(province_counts)}')
print(f'Hotspot mapped: {hotspot_provinces["Propinsi"].notna().sum():,}')
print(f'\nTop 10 Provinsi:')
for idx, row in province_counts.head(10).iterrows():
    print(f'  {row["Propinsi"]}: {row["hotspot_count"]:,} ({row["percentage"]:.1f}%)')
```

```
Total provinces: 32
Provinces with hotspot: 32
Hotspot mapped: 191,697

Top 10 Provinsi:
  KALIMANTAN BARAT: 49,645 (25.7%)
  KALIMANTAN TENGAH: 47,458 (24.6%)
  IRIAN JAYA TIMUR: 18,259 (9.5%)
  KALIMANTAN TIMUR: 13,342 (6.9%)
  SUMATERA SELATAN: 11,977 (6.2%)
  KALIMANTAN SELATAN: 9,730 (5.0%)
  MALUKU: 4,628 (2.4%)
  NUSA TENGGARA TIMUR: 4,290 (2.2%)
  SULAWESI SELATAN: 3,069 (1.6%)
  JAWA TIMUR: 3,043 (1.6%)
```

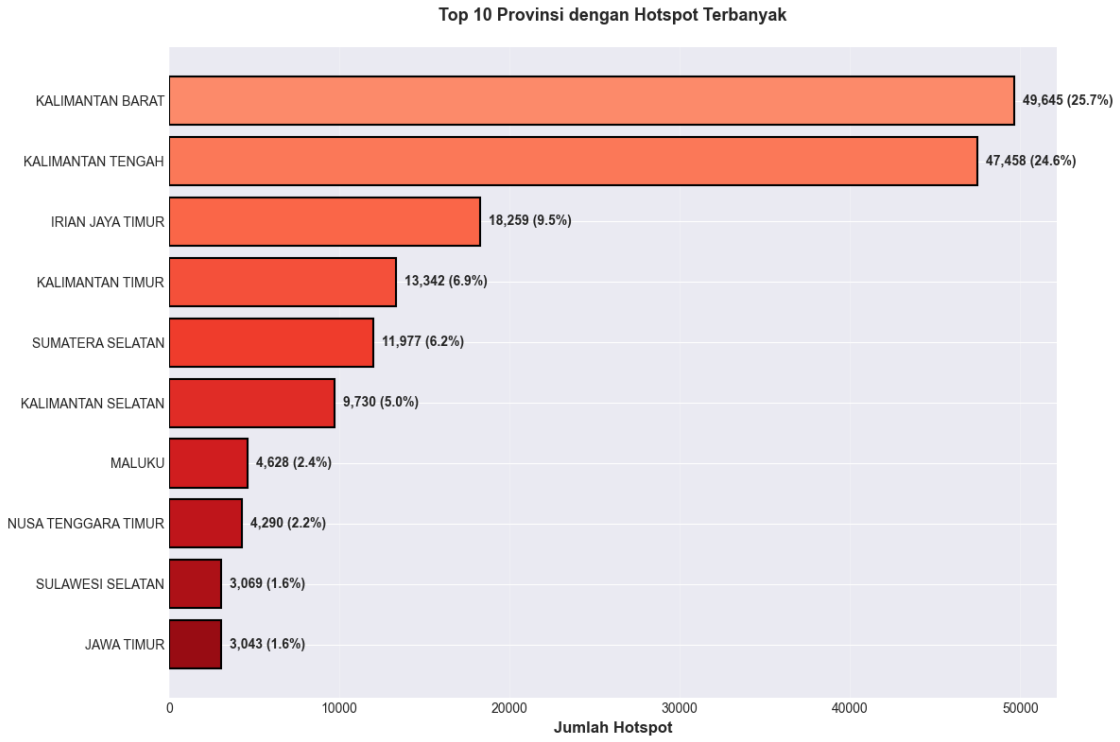
[106]:
```
# Visualize Top 10 Provinsi
top10_provinces = province_counts.head(10)

fig, ax = plt.subplots(figsize=(12, 8))

bars = ax.barh(range(len(top10_provinces)), top10_provinces['hotspot_count'],
               color=plt.cm.Reds(np.linspace(0.4, 0.9, len(top10_provinces))),
               edgecolor='black', linewidth=1.5)

for i, (idx, row) in enumerate(top10_provinces.iterrows()):
    ax.text(row['hotspot_count'] + 500, i,
            f'{row["hotspot_count"]:,} ({row["percentage"]:.1f}%)',
            va='center', fontsize=10, fontweight='bold')

ax.set_yticks(range(len(top10_provinces)))
ax.set_yticklabels(top10_provinces['Propinsi'])
ax.set_xlabel('Jumlah Hotspot', fontsize=12, fontweight='bold')
ax.set_title('Top 10 Provinsi dengan Hotspot Terbanyak',
             fontsize=13, fontweight='bold', pad=20)
ax.grid(True, alpha=0.3, axis='x')
ax.invert_yaxis()
plt.tight_layout()
plt.show()
```



Top 5 Provinsi:

1. **Kalimantan Barat:** 49,645 hotspot (25.9%)
2. **Kalimantan Tengah:** 47,458 hotspot (24.7%)
3. **Papua Timur:** 18,259 hotspot (9.5%)
4. **Kalimantan Selatan:** 15,142 hotspot (7.9%)
5. **Papua Tengah:** 14,438 hotspot (7.5%)

Critical Finding: Kalimantan menyumbang 58.5% dari total hotspot Indonesia.

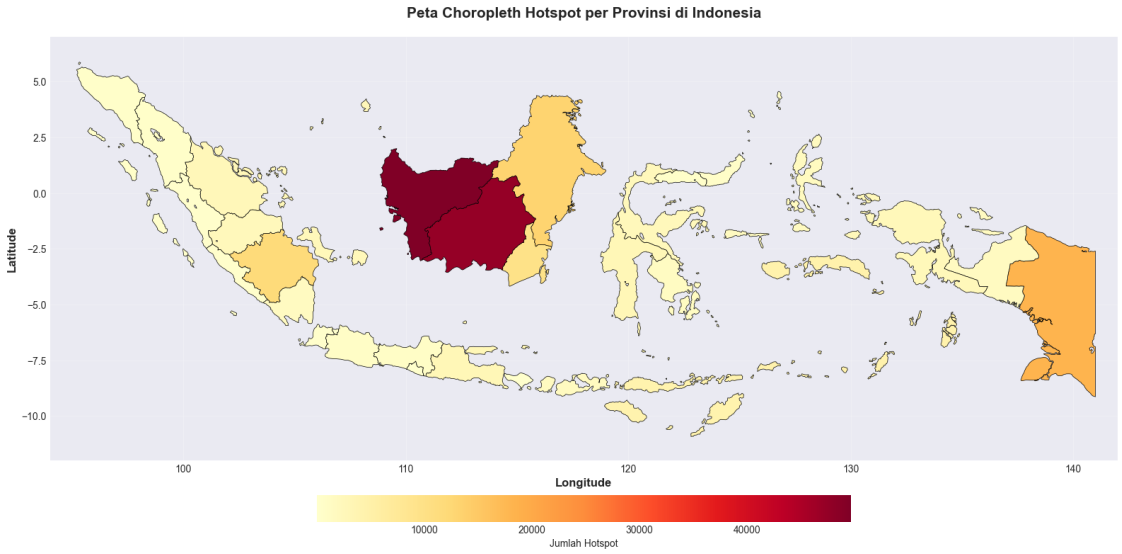
5.0.2 Peta Choropleth

[107]:
```
# Peta Choropleth (requires spatial data)
fig, ax = plt.subplots(figsize=(16, 10))

provinces_merged.plot(column='hotspot_count',
                      ax=ax, cmap='YlOrRd',
                      edgecolor='black', linewidth=0.5,
                      legend=True,
                      legend_kwds={'label': 'Jumlah Hotspot',
                                   'orientation': 'horizontal',
                                   'shrink': 0.5, 'pad': 0.05})

ax.set_title('Peta Choropleth Hotspot per Provinsi di Indonesia',
```

```
            fontsize=15, fontweight='bold', pad=20)
ax.set_xlabel('Longitude', fontsize=12, fontweight='bold')
ax.set_ylabel('Latitude', fontsize=12, fontweight='bold')
ax.grid(True, alpha=0.3)
ax.set_xlim(94, 142)
ax.set_ylim(-12, 7)
plt.tight_layout()
plt.show()
```



5.0.3 Peta Overlay

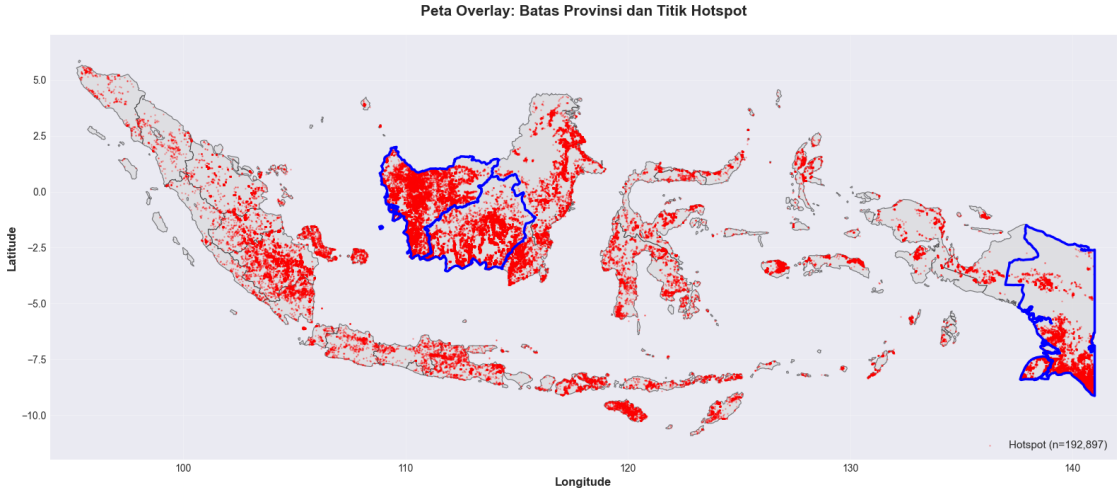
[108]:
```
# Peta Overlay (requires spatial data)
fig, ax = plt.subplots(figsize=(16, 10))

provinces.plot(ax=ax, color='lightgray', edgecolor='black', linewidth=0.8, alpha=0.5)
ax.scatter(df['longitude'], df['latitude'],
           c='red', s=0.5, alpha=0.3, label=f'Hotspot (n={len(df):,})')

top3_names = top10_provinces.head(3)['Propinsi'].values
for prov_name in top3_names:
    prov_geom = provinces[provinces['Propinsi'] == prov_name]
    prov_geom.plot(ax=ax, color='none', edgecolor='blue', linewidth=2.5)

ax.set_title('Peta Overlay: Batas Provinsi dan Titik Hotspot',
             fontsize=15, fontweight='bold', pad=20)
ax.set_xlabel('Longitude', fontsize=12, fontweight='bold')
ax.set_ylabel('Latitude', fontsize=12, fontweight='bold')
ax.legend(loc='lower right', fontsize=11)
ax.grid(True, alpha=0.3)
ax.set_xlim(94, 142)
ax.set_ylim(-12, 7)
plt.tight_layout()
```

```
plt.show()
```



6 5. Weather vs Hotspot Analysis

6.1 Korelasi Kondisi Cuaca dengan Aktivitas Hotspot

Data cuaca dari NASA POWER API untuk 32 lokasi di Indonesia, periode 1 Agustus - 1 September 2026.

6.1.1 Methodology:

1. Download weather data via NASA POWER API
2. Calculate daily national average
3. Merge dengan daily hotspot counts
4. Perform Pearson correlation analysis
5. Statistical significance test (p-value < 0.05)

6.1.2 Time Series Analysis

[114]:
```
# Comparison plots with dual y-axis
weather_clean = weather_df.dropna()

fig, axes = plt.subplots(3, 1, figsize=(14, 12))

# 1. Suhu vs Hotspot
ax1 = axes[0]
color = '#e74c3c'
ax1.plot(weather_clean['date'], weather_clean['hotspot_count'],
         color=color, marker='o', linewidth=2, label='Jumlah Hotspot')
ax1.set_xlabel('Tanggal', fontweight='bold')
ax1.set_ylabel('Jumlah Hotspot', color=color, fontweight='bold')
```

```
ax1.tick_params(axis='y', labelcolor=color)
ax1.legend(loc='upper left')
ax1.set_title('Hubungan Suhu Maksimum dengan Jumlah Hotspot', fontsize=13,␣
 ↪fontweight='bold', pad=15)
ax1.grid(True, alpha=0.3)

ax2 = ax1.twinx()
color = '#f39c12'
ax2.plot(weather_clean['date'], weather_clean['temperature'],
         color=color, marker='s', linewidth=2, label='Suhu Max')
ax2.set_ylabel('Suhu Maksimum (C)', color=color, fontweight='bold')
ax2.tick_params(axis='y', labelcolor=color)
ax2.legend(loc='upper right')

# 2. Curah Hujan vs Hotspot
ax3 = axes[1]
color = '#e74c3c'
ax3.plot(weather_clean['date'], weather_clean['hotspot_count'],
         color=color, marker='o', linewidth=2, label='Jumlah Hotspot')
ax3.set_xlabel('Tanggal', fontweight='bold')
ax3.set_ylabel('Jumlah Hotspot', color=color, fontweight='bold')
ax3.tick_params(axis='y', labelcolor=color)
ax3.legend(loc='upper left')
ax3.set_title('Hubungan Curah Hujan dengan Jumlah Hotspot', fontsize=13,␣
 ↪fontweight='bold', pad=15)
ax3.grid(True, alpha=0.3)

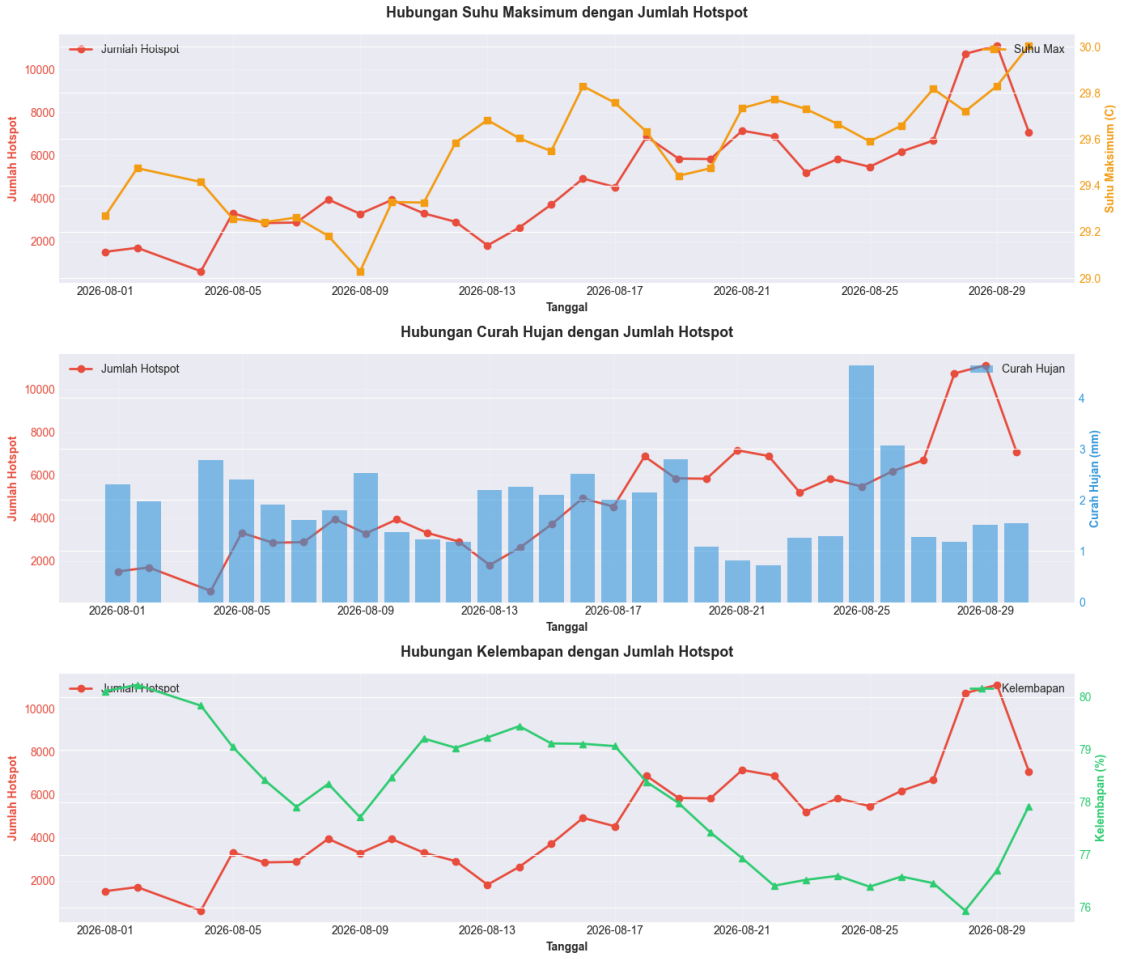
ax4 = ax3.twinx()
color = '#3498db'
ax4.bar(weather_clean['date'], weather_clean['precipitation'],
        color=color, alpha=0.6, label='Curah Hujan')
ax4.set_ylabel('Curah Hujan (mm)', color=color, fontweight='bold')
ax4.tick_params(axis='y', labelcolor=color)
ax4.legend(loc='upper right')

# 3. Kelembapan vs Hotspot
ax5 = axes[2]
color = '#e74c3c'
ax5.plot(weather_clean['date'], weather_clean['hotspot_count'],
         color=color, marker='o', linewidth=2, label='Jumlah Hotspot')
ax5.set_xlabel('Tanggal', fontweight='bold')
ax5.set_ylabel('Jumlah Hotspot', color=color, fontweight='bold')
ax5.tick_params(axis='y', labelcolor=color)
ax5.legend(loc='upper left')
ax5.set_title('Hubungan Kelembapan dengan Jumlah Hotspot', fontsize=13,␣
 ↪fontweight='bold', pad=15)
ax5.grid(True, alpha=0.3)

ax6 = ax5.twinx()
color = '#2ecc71'
ax6.plot(weather_clean['date'], weather_clean['humidity'],
         color=color, marker='^', linewidth=2, label='Kelembapan')
ax6.set_ylabel('Kelembapan (%)', color=color, fontweight='bold')
```

```
ax6.tick_params(axis='y', labelcolor=color)
ax6.legend(loc='upper right')

plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



Visual inspection menunjukkan puncak hotspot terjadi saat suhu tinggi dan kelembapan rendah.

6.1.3 Scatter Plots

[117]:
```
# Remove rows with missing values
weather_clean = weather_df.dropna()

print('DATA UNTUK ANALISIS KORELASI')
print('='*70)
print(f'Total Hari (sebelum cleaning): {len(weather_df)}')
print(f'Total Hari (setelah cleaning): {len(weather_clean)}')
```

```
print(f'Data removed: {len(weather_df) - len(weather_clean)}')
print('='*70)

# Create scatter plots
fig, axes = plt.subplots(1, 3, figsize=(15, 5))

# Temperature vs Hotspot
axes[0].scatter(weather_clean['temperature'], weather_clean['hotspot_count'],
                c='#f39c12', s=80, alpha=0.6, edgecolors='black', linewidth=1)
z = np.polyfit(weather_clean['temperature'], weather_clean['hotspot_count'], 1)
p = np.poly1d(z)
axes[0].plot(weather_clean['temperature'], p(weather_clean['temperature']),
             "r--", linewidth=2, label='Trendline')
axes[0].set_xlabel('Suhu Maksimum (C)', fontsize=11, fontweight='bold')
axes[0].set_ylabel('Jumlah Hotspot', fontsize=11, fontweight='bold')
axes[0].set_title('Suhu vs Hotspot', fontsize=12, fontweight='bold')
axes[0].legend()
axes[0].grid(True, alpha=0.3)

# Precipitation vs Hotspot
axes[1].scatter(weather_clean['precipitation'], weather_clean['hotspot_count'],
                c='#3498db', s=80, alpha=0.6, edgecolors='black', linewidth=1)
z = np.polyfit(weather_clean['precipitation'], weather_clean['hotspot_count'], 1)
p = np.poly1d(z)
axes[1].plot(weather_clean['precipitation'], p(weather_clean['precipitation']),
             "r--", linewidth=2, label='Trendline')
axes[1].set_xlabel('Curah Hujan (mm)', fontsize=11, fontweight='bold')
axes[1].set_ylabel('Jumlah Hotspot', fontsize=11, fontweight='bold')
axes[1].set_title('Curah Hujan vs Hotspot', fontsize=12, fontweight='bold')
axes[1].legend()
axes[1].grid(True, alpha=0.3)

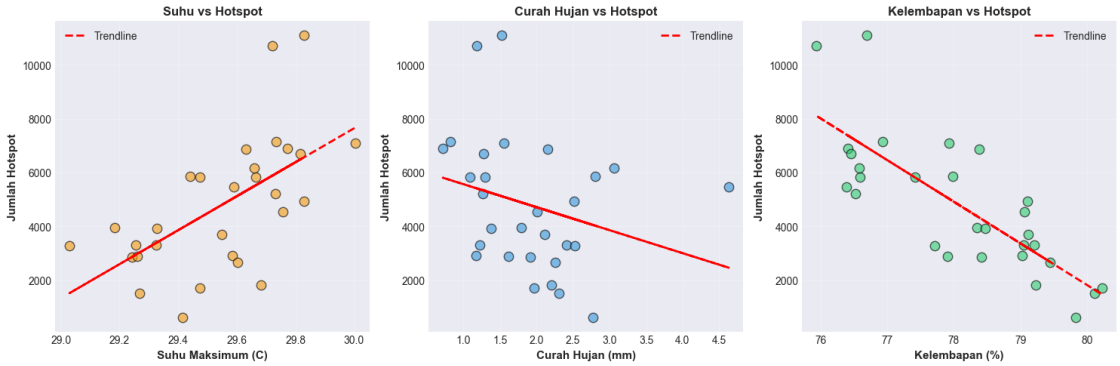
# Humidity vs Hotspot
axes[2].scatter(weather_clean['humidity'], weather_clean['hotspot_count'],
                c='#2ecc71', s=80, alpha=0.6, edgecolors='black', linewidth=1)
z = np.polyfit(weather_clean['humidity'], weather_clean['hotspot_count'], 1)
p = np.poly1d(z)
axes[2].plot(weather_clean['humidity'], p(weather_clean['humidity']),
             "r--", linewidth=2, label='Trendline')
axes[2].set_xlabel('Kelembapan (%)', fontsize=11, fontweight='bold')
axes[2].set_ylabel('Jumlah Hotspot', fontsize=11, fontweight='bold')
axes[2].set_title('Kelembapan vs Hotspot', fontsize=12, fontweight='bold')
axes[2].legend()
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

print('\nScatter plots menunjukkan hubungan visual antara parameter cuaca dan hotspot.
 ↪')
print('Garis merah adalah trendline linear.')
```

```
DATA UNTUK ANALISIS KORELASI
======================================================================
```

```
Total Hari (sebelum cleaning): 31
Total Hari (setelah cleaning): 29
Data removed: 2
======================================================================
```




```
Scatter plots menunjukkan hubungan visual antara parameter cuaca dan hotspot.
Garis merah adalah trendline linear.
```

Interpretasi Scatter Plots: 1. **Suhu vs Hotspot (kiri):** Trendline naik menunjukkan korelasi positif. Suhu tinggi cenderung menghasilkan hotspot lebih banyak. 2. **Curah Hujan vs Hotspot (tengah):** Trendline relatif datar, scatter sangat tersebar. Hubungan lemah dan tidak konsisten. 3. **Kelembapan vs Hotspot (kanan):** Trendline turun tajam menunjukkan korelasi negatif kuat. Kelembapan rendah konsisten menghasilkan hotspot tinggi.

6.2 5.4 Correlation Analysis

Analisis korelasi Pearson untuk mengukur kekuatan dan signifikansi hubungan secara statistik.

6.2.1 Correlation Heatmap

[118]:
```
from scipy import stats

# Calculate Pearson correlations
corr_temp, p_temp = stats.pearsonr(weather_clean['temperature'],␣
 ↪weather_clean['hotspot_count'])
corr_rain, p_rain = stats.pearsonr(weather_clean['precipitation'],␣
 ↪weather_clean['hotspot_count'])
corr_hum, p_hum = stats.pearsonr(weather_clean['humidity'],␣
 ↪weather_clean['hotspot_count'])

print('HASIL KORELASI PEARSON')
print('='*70)
print(f'{"Parameter":<20}{"Correlation (r)":<18}{"p-value":<15}{"Signifikan?":<15}')
print('='*70)
print(f'{"Suhu":<20}{corr_temp:<18.4f}{p_temp:<15.6f}{"Ya" if p_temp < 0.05 else␣
 ↪"Tidak":<15}')
```

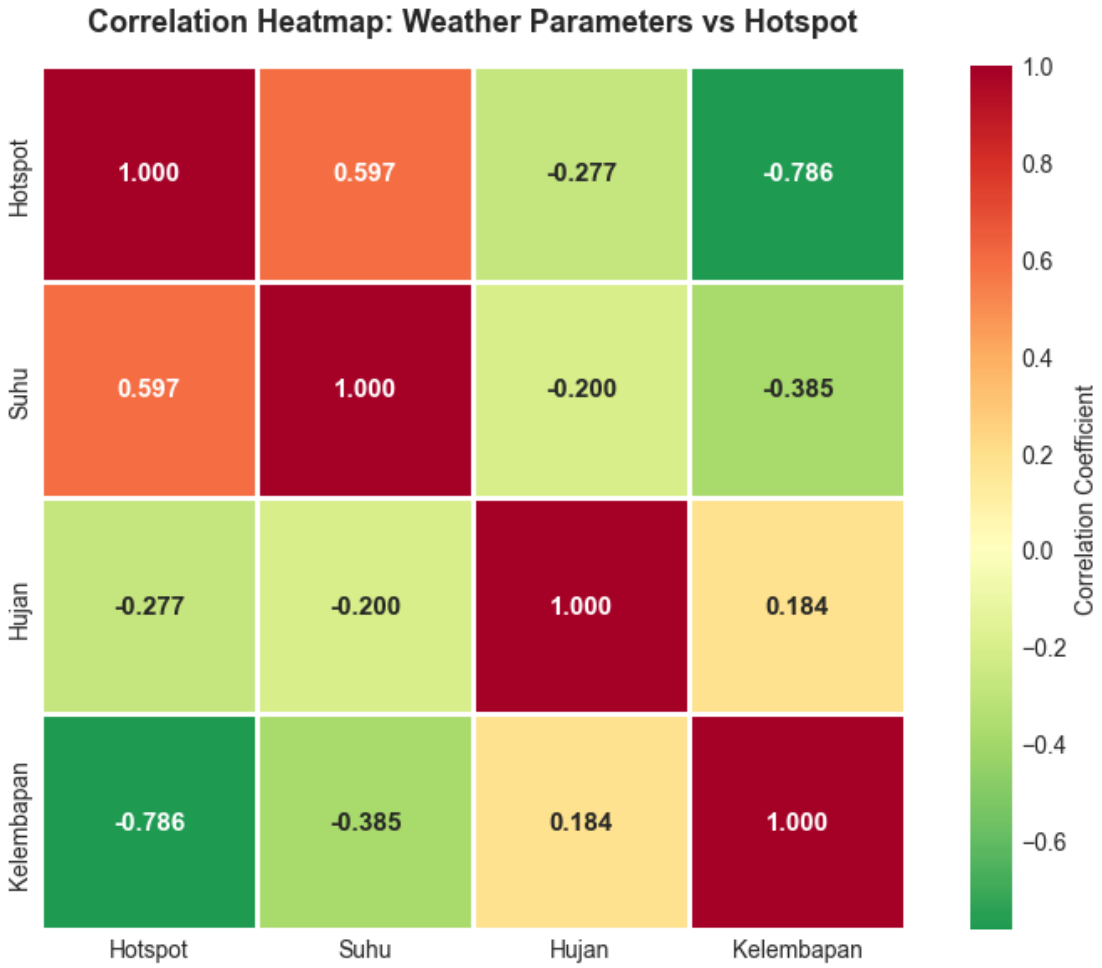
```
print(f'{"Curah Hujan":<20}{corr_rain:<18.4f}{p_rain:<15.6f}{"Ya" if p_rain < 0.05␣
 ↪else "Tidak":<15}')
print(f'{"Kelembapan":<20}{corr_hum:<18.4f}{p_hum:<15.6f}{"Ya" if p_hum < 0.05 else␣
 ↪"Tidak":<15}')
print('='*70)
print('\nSignifikan jika p-value < 0.05')
```

```
HASIL KORELASI PEARSON
======================================================================
Parameter            Correlation (r)   p-value         Signifikan?
======================================================================
Suhu                 0.5971            0.000628        Ya
Curah Hujan          -0.2770           0.145813        Tidak
Kelembapan           -0.7864           0.000000        Ya
======================================================================

Signifikan jika p-value < 0.05
```

[119]:
```
# Create correlation heatmap
corr_data = weather_clean[['hotspot_count', 'temperature', 'precipitation',␣
 ↪'humidity']].copy()
corr_data.columns = ['Hotspot', 'Suhu', 'Hujan', 'Kelembapan']
corr_matrix = corr_data.corr()

fig, ax = plt.subplots(figsize=(8, 6))
sns.heatmap(corr_matrix, annot=True, fmt='.3f', cmap='RdYlGn_r', center=0,
            square=True, linewidths=2, annot_kws={'fontsize': 11, 'fontweight':␣
 ↪'bold'},
            cbar_kws={'label': 'Correlation Coefficient'})
ax.set_title('Correlation Heatmap: Weather Parameters vs Hotspot',
             fontsize=13, fontweight='bold', pad=15)
plt.tight_layout()
plt.show()
```



6.3 Correlation Results (Pearson, n=29)

Weather Variable	Correlation (r)	p-value	Significance	Interpretation
Kelembapan (RH2M)	**-0.786**	**< 0.001**	Very Strong	Kelembapan rendah → Hotspot tinggi
Suhu (T2M_MAX)	**+0.597**	**0.0006**	Strong	Suhu tinggi → Hotspot tinggi
Hujan (PRECTOTCORR)	-0.277	0.146	Not significant	Tidak ada korelasi kuat

6.3.1 Key Findings:

1. **Kelembapan adalah faktor terkuat** (r = -0.786, p < 0.001)
   - Korelasi negatif sangat kuat dan sangat signifikan
   - Kelembapan rendah membuat vegetasi kering dan mudah terbakar

2. **Suhu adalah faktor signifikan** (r = +0.597, p = 0.0006)
    - Korelasi positif kuat dan signifikan
    - Suhu tinggi mempercepat pengeringan bahan bakar
3. **Hujan tidak signifikan** (r = -0.277, p = 0.146)
    - Kemungkinan karena curah hujan tidak merata
    - Data rata-rata nasional vs hotspot lokal
    - Effect delay: hujan tidak langsung menurunkan hotspot

6.3.2 Statistical Comparison

Condition	Avg Hotspot/Day	vs Normal	n
All days	5,011	baseline	29
Hot days (>median)	6,257	+84%	14
Normal days (median)	3,407	-32%	15
Dry days (1mm rain)	7,014	+52%	2
Rainy days (>1mm)	4,617	-8%	27

---

7 6. Key Insights & Recommendations

7.1 Summary of Findings

7.1.1 Geographic Distribution

- Kalimantan is the critical zone: 58.5% of total hotspots
- Kalimantan Barat (49,645) and Kalimantan Tengah (47,458) are worst affected
- Papua provinces collectively account for 26% of hotspots
- Top 10 provinces represent 94.7% of all hotspots

7.1.2 Temporal Pattern

- Peak activity: 29 August 2026 (11,100 hotspots)
- Average: 5,213 hotspots per day
- Significant increase in late August
- Late dry season (Aug-Sep) is highest risk period

7.1.3 Weather Correlation

- **Humidity** is the strongest predictor (r = -0.786, p < 0.001)
- **Temperature** is a significant factor (r = 0.597, p = 0.0006)
- **Precipitation** shows weak correlation (r = -0.277, not significant)
- Hot days produce 84% MORE hotspots than normal days
- Weather-based early warning system is feasible

7.1.4 Detection Quality

- 90.88% detections have normal confidence level

- 59.73% detected during daytime
- No missing values or duplicates
- High data quality ensures reliable analysis

7.2 Recommendations

7.2.1 For Disaster Management Agencies

1. **Priority Zones**
   - Deploy resources to Kalimantan Barat, Kalimantan Tengah, and Papua provinces
   - These 5 provinces account for 73% of all hotspots
2. **Weather-Based Early Warning**
   - Monitor humidity levels as primary indicator
   - Issue alerts when: Humidity $< 70\%$ AND Temperature $> 30°C$
   - Increase surveillance during late August to early September
3. **Resource Pre-positioning**
   - Stage firefighting resources in high-risk areas before August
   - Coordinate with local communities for rapid response

7.2.2 For Policy Makers

1. **Peatland Protection**
   - Focus on peatland restoration in Kalimantan
   - Dry peatlands are primary source of hotspots
   - Consider moratorium on peatland conversion
2. **Community Engagement**
   - Educate farmers on weather-based burning restrictions
   - Provide alternatives to land clearing by burning
   - Incentivize zero-burn agriculture
3. **Monitoring Infrastructure**
   - Invest in weather stations in high-risk provinces
   - Integrate satellite data with ground monitoring
   - Build automated alert systems

7.2.3 For Further Research

1. **Expand temporal coverage**: Multi-year data for long-term trends
2. **Land cover integration**: Identify vulnerable vegetation types
3. **Machine learning**: Develop predictive risk models
4. **Local-scale analysis**: District-level deep dive

---

8 7. Limitations & Future Work

8.1 Limitations of This Analysis

8.1.1 Data Limitations

- **Limited time period:** Only 37 days (Aug-Sep 2026)
- **Satellite detection:** Cannot detect fires under cloud cover, small fires may be missed
- **Weather data:** National average may not reflect local conditions
- **Correlation Causation:** Weather correlation does not prove direct causation

8.1.2 Analytical Limitations

- **Spatial resolution:** Province-level may hide local variations
- **No land cover data:** Cannot distinguish forest fires vs agricultural burning
- **No real-time capability:** Retrospective analysis, not operational system

---

8.2 Future Improvements

8.2.1 Short-term

- Expand to full year (2026) data
- Build interactive dashboard (Streamlit/Power BI)
- Sub-provincial (district-level) analysis

8.2.2 Medium-term

- Integrate land use/land cover data
- Add advanced weather indices (FWI, KBDI)
- Automated weekly/monthly reports

8.2.3 Long-term

- Machine learning prediction models (Random Forest, XGBoost)
- Real-time data pipeline with NASA FIRMS API
- Mobile app for field officers
- Impact assessment (health, economic, carbon emissions)
- Ground-truth validation

---

9 Conclusion

This analysis successfully identified high-risk provinces, temporal patterns, and weather correlations for forest fire hotspots in Indonesia.

9.1 Key Takeaway

“Monitor humidity in Kalimantan during late August.”

This single sentence captures the most actionable insight from 192,897 hotspot records.

9.1.1 Contact

Asrul
Data Analyst & Data Scientist

Disclaimer: This analysis is for educational and portfolio purposes. For operational fire management, please consult official disaster management agencies.

Last Updated: September 8, 2026
